MONET – Supplementary Material

Benchmark Environments and Closed-Form Task Fitness Definitions

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This document accompanies the MONET paper *"Multi-Task Optimization over Networks of Tasks"* (Hatzky et al., 2026) and the code repository at https://github.com/ju2ez/MONET. It collects the per-environment task / solution parameterisations and the closed-form fitness expressions used in the four benchmark domains.

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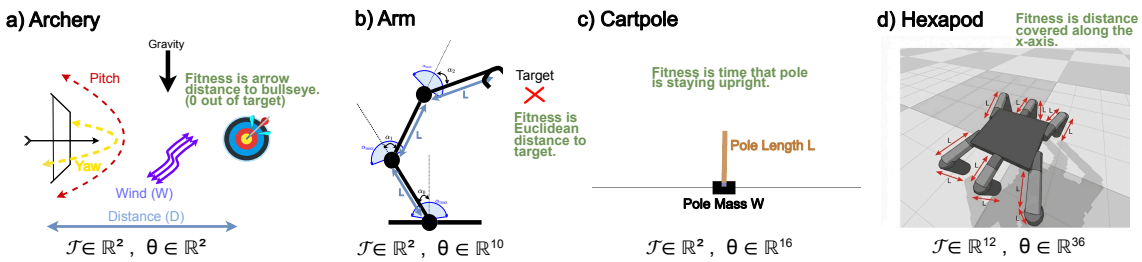

Figure 1: Overview of benchmark environments used to evaluate MONET: (a) 10-DoF Arm, (b) Archery, (c) Cartpole with variable morphology, and (d) Six-legged locomotion (Hexapod).

All four benchmarks define fitness on $[0, 1]$ (higher is better) and are run for 10^6 evaluations with a batch size of 64, replicated over 20 random seeds. Task parameters $\boldsymbol{\tau} \in \Theta$ define the family of problems; solution vectors $\mathbf{x} \in \mathcal{X}$ are the decision variables optimised per task.

1 Benchmark Environments

1.1 Kinematic Arm with Variable Morphology (10-DoF-Arm)

A planar 10-DoF arm must place its end-effector at a fixed target (Figure 1a); adapted from Mouret & Maguire (2020).

Tasks Θ maximum joint angle $\alpha_{\max}$ and link length L.

Solutions $\mathcal{X}$ 10 joint angles, normalised by $\alpha_{\max}$.

Fitness $\exp\bigl(-\|\mathbf{p}_{\text{ee}} - \mathbf{p}_{\text{tar}}\|^2\bigr)$.

Number of tasks 5,000.

1.2 Archery with Adaptive Environment

A projectile-aiming task in which the archer compensates for distance and wind (Figure 1b); adapted from Anne & Mouret (2024).

Tasks Θ target distance $D \in [5, 40]\,\text{m}$ and horizontal wind $W \in [-10, 10]\,\text{m/s}^2$.

Solutions $\mathcal{X}$ arrow yaw and pitch, each in $[-\pi/12, \pi/12]$ at fixed launch speed $v_0 = 70\,\text{m/s}$.

Fitness normalised archery score derived from the analytic miss distance on the target plane.

Number of tasks 5,000.

1.3 Cartpole with Variable Morphology

Classic cart-pole balancing under varying physical parameters (Figure 1c); Michie & Chambers (1968), Barto et al. (1983).

Tasks Θ pole length $L \in [0.3, 1.2]\,\text{m}$, pole mass $m_p \in [0.05, 0.5]\,\text{kg}$, cart mass $m_c \in [0.2, 2.0]\,\text{kg}$.

Solutions $\mathcal{X}$ control policy mapping state $(x_c, \dot{x}_c, \phi, \dot{\phi})$ to control force F.

Fitness fraction of the episode during which the pole stays within $|\phi| < 12^\circ$ and $|x_c| < 2.4\,\text{m}$.

Number of tasks 5,000.

1.4 Six-Legged Locomotion (Hexapod)

A simulated hexapod must walk forward across varying body morphologies (Figure 1d), simulated in PyBullet (Coumans & Bai, 2016–2021); adapted from Cully et al. (2015) and Mouret & Maguire (2020).

Tasks Θ the 12 leg-segment lengths (2 000 random morphologies).

Solutions $\mathcal{X}$ a 36-D gait vector encoding phase, amplitude, and offset of a smoothed square-wave trajectory for each of 12 controlled joints; the remaining 6 joints mirror their neighbours to keep the distal segment vertical.

Fitness forward displacement ΔX over a 3 s episode, normalised to $[0, 1]$; episodes terminate early if body pitch/roll exceeds $\pi/8$ or lateral drift exceeds 0.5 m.

Number of tasks 2,000.

2 Closed-Form Task Fitness Definitions

This appendix gives the explicit fitness expressions for the four benchmarks summarised in Section 1.

2.1 10-DoF Arm

With end-effector position $\mathbf{p}_{\text{ee}}$ and target $\mathbf{p}_{\text{tar}}$,

$$f = \exp\bigl(-\|\mathbf{p}_{\text{ee}} - \mathbf{p}_{\text{tar}}\|^2\bigr) \in [0, 1].$$

2.2 Archery

The initial velocity vector is

$$\mathbf{v} = v_0 \begin{bmatrix} -\sin(\text{yaw}) \\ \cos(\text{yaw})\cos(\text{pitch}) \\ \cos(\text{yaw})\sin(\text{pitch}) \end{bmatrix}, \qquad t_{\text{hit}} = \frac{D}{v_y},$$

and, with g_z the gravitational acceleration along z and W_x the wind acceleration along x, the miss distance on the target plane is

$$r = \left\| \tfrac{1}{2}(-g_z\,\hat{\mathbf{e}}_z + W_x\,\hat{\mathbf{e}}_x)\,t_{\text{hit}}^2 + \mathbf{v}\,t_{\text{hit}} \right\|_2, \qquad f = \frac{\max\!\left(0,\, \text{int}\!\left(10 - \frac{r}{0.061}\right)\right)}{10}.$$

2.3 Cartpole

With pole angle ϕ, cart position x_c, control force F, and gravity g,

$$\ddot{\phi} = \frac{g\sin\phi + \cos\phi\left(-F - m_p L\dot{\phi}^2\sin\phi\right)/(m_c + m_p)}{L\left(\frac{4}{3} - \frac{m_p\cos^2\phi}{m_c+m_p}\right)},$$

$$\ddot{x}_c = \frac{F + m_p L\left(\dot{\phi}^2\sin\phi - \ddot{\phi}\cos\phi\right)}{m_c + m_p}.$$

Fitness is $f = t_{\text{balanced}}/t_{\max}$, the fraction of the episode the pole stays within $|\phi| < 12^\circ$ and $|x_c| < 2.4\,\text{m}$.

2.4 Hexapod

With gait parameters $\mathbf{x} \in \mathcal{X}$ and morphology $\boldsymbol{\tau} \in \Theta$,

$$f(\mathbf{x}, \boldsymbol{\tau}) = \Delta X,$$

the forward displacement of the centre of mass over a 3 s episode (normalised to $[0, 1]$), with early termination if body pitch or roll exceeds $\pi/8$ or $|Y| > 0.5\,\text{m}$.

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